

Architecture Selection Justification

We chose Layered Architecture for the Blood Bank & Donation Management System.

Architectural Choice

The system is organized into three layers: a Presentation Layer (a single User Interface Component serving the Donor, Hospital, Staff, and Admin pages described in SRS section 3.1), a Business Layer (Authentication & RBAC, Donor Management, Inventory Management, Hospital Request, and Drive Management Components, corresponding directly to the five feature areas in SRS section 4), and a Data Layer (a single Database Component). An external Notification Service, consistent with SRS section 3.3 and the system's own use-case diagrams in section 7, is accessed by the Authentication and Drive Management components for OTP delivery and slot reminders.

Two Reasons for This Choice

1. The five feature areas map cleanly onto layers: Authentication/RBAC, Donor Registration, Inventory Tracking, Hospital Requests, and Drive Management (sections 4.1-4.5) are naturally distinct business-layer components sitting behind one shared presentation layer and one shared data layer. As stated in SRS section 2.1, this is deliberately "a simple web application" for a single organisation, not a system that needs independently deployed or independently scaled services, so the added operational complexity of Microservices is not justified here.
2. BBS-NF005 (preventing two staff members from allocating the same blood unit at the same time) is far easier to guarantee correctly with a single Business Layer coordinating access to one Database Component than across independently deployed services, where the same guarantee would require distributed locking or a saga pattern just to achieve what a single database transaction handles natively in a layered design.

Security Advantage

Because the Data Layer is only reachable through the Business Layer, role-based access control (BBS-SR003: a Donor account must not reach Staff/Admin-only pages or actions) is enforced centrally in the Authentication & RBAC Component rather than being duplicated or potentially bypassed across separate services. This same centralisation also makes it straightforward to satisfy BBS-SR004, since key actions such as approving a request or marking a unit expired pass through one Business Layer where they can be logged consistently with user and timestamp.

Performance Benefit

BBS-NF001 requires inventory search results to load in under 1-2 seconds. With a Layered Architecture, the Inventory Management Component has direct, low-latency access to the Database Component without the network hops a Microservices approach would introduce between separate inventory and search services. This directness also supports BBS-NF005's concurrency guarantee, since allocation checks and inventory reads pass through the same in-process Business Layer rather than being coordinated across a network boundary.